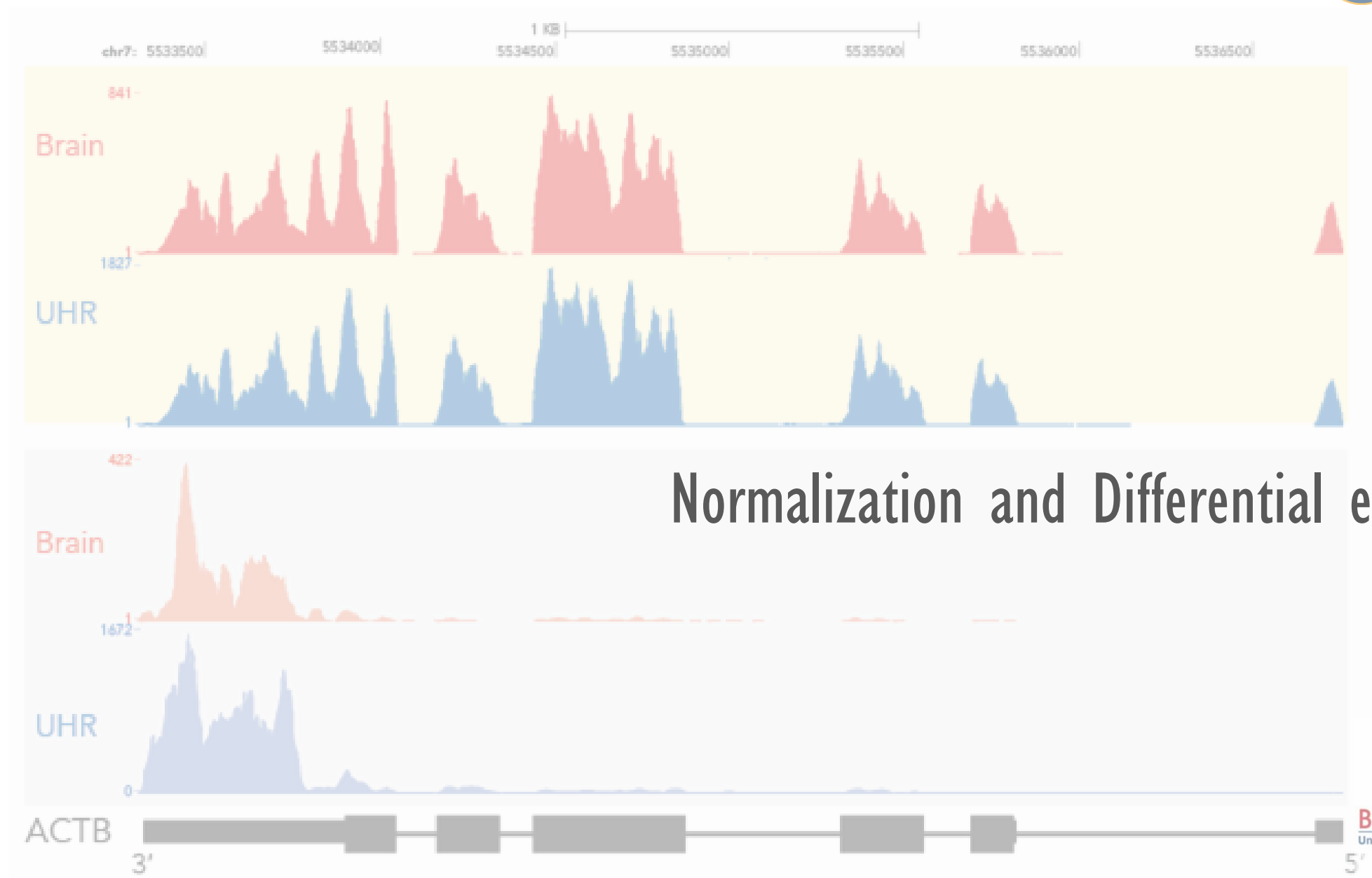




Why do we care about normalization?

Why do we care about normalization?

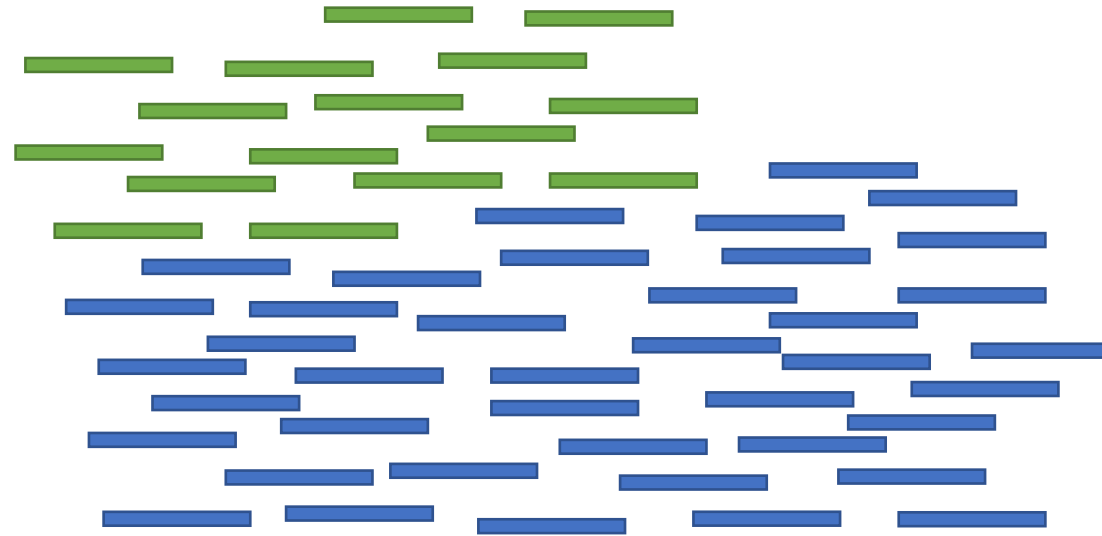
- Every sequence represents a part of an RNA molecule present in the sample
- Once the mapping is done, the read count gives information about the abundance of each gene expressed in our sample
- However, the read count needs to be adjusted to be comparable between isoforms, genes, samples and experiments.

Normalization is the process of scaling raw counts values to make them comparable

Why do we care about normalization?

Given the number of reads in the draw, **which isoform is more expressed?**

Reads



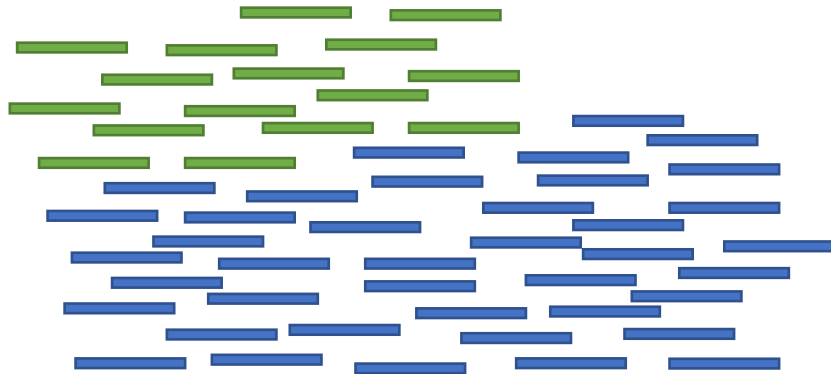
Isoforms



Why do we care about normalization?

Given the number of reads in the draw, **which isoform is more expressed?**

Reads



Isoforms



The blue isoform **has more reads, but it is also longer than the green one.**

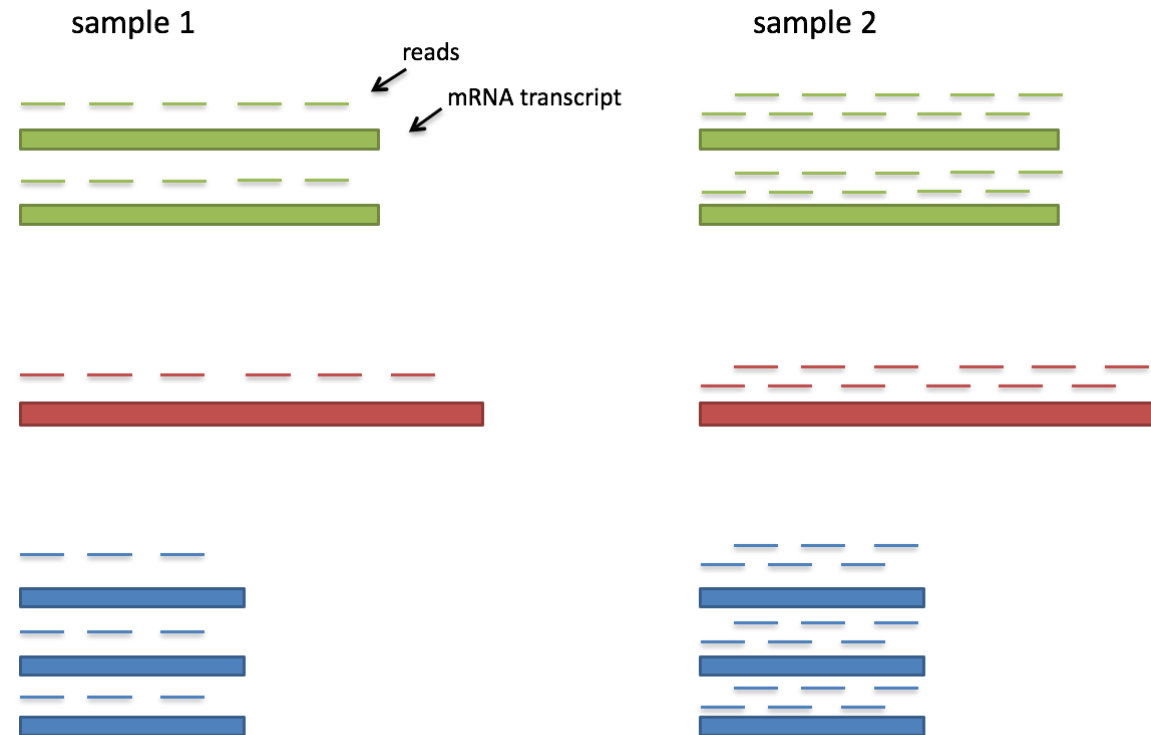
As the library preparation includes fragmentation, long isoforms will generate a greater number of reads

Principal factors included on the normalization

- Global Sequence Coverage
 - Genes Length
- Transcriptome composition

Global Sequence Coverage

In this case, it is the total number of mapped reads to the genome or transcriptome. This is very important to normalize when comparing expressions **between samples**.



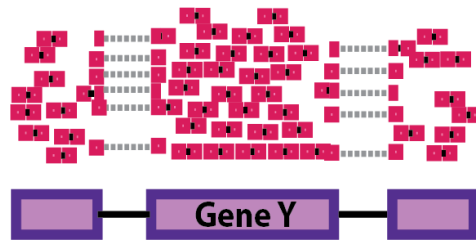
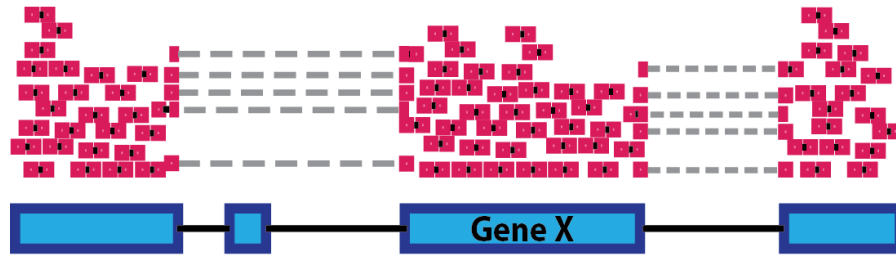
MI Love: RNA-seq statistical analysis

In sample 2, all genes seem to be expressed twice. However, the sample 2 was sequenced on double coverage...so the differences we see in the sample 2 is just an artefact of the sequencing.

Gene Length

When comparing expression **between genes on the same samples** (let's say comparing the expression of gene A with gene B) the length will tend to overestimate the expression of long genes.

Sample A Reads



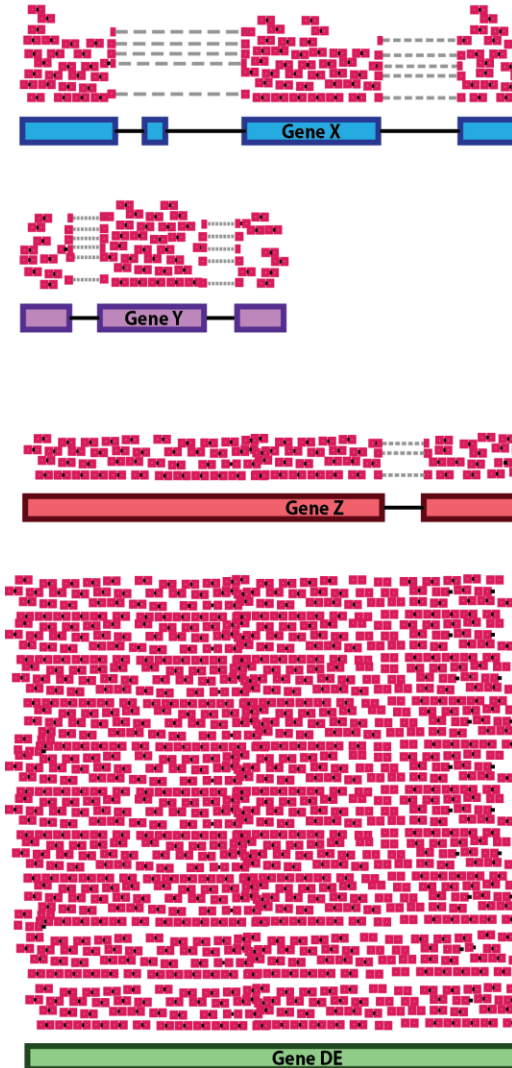
Indeed, the number of mapped reads to a long gene might be the same as that of the short gene, in this case, the former is more expressed

Transcriptome composition

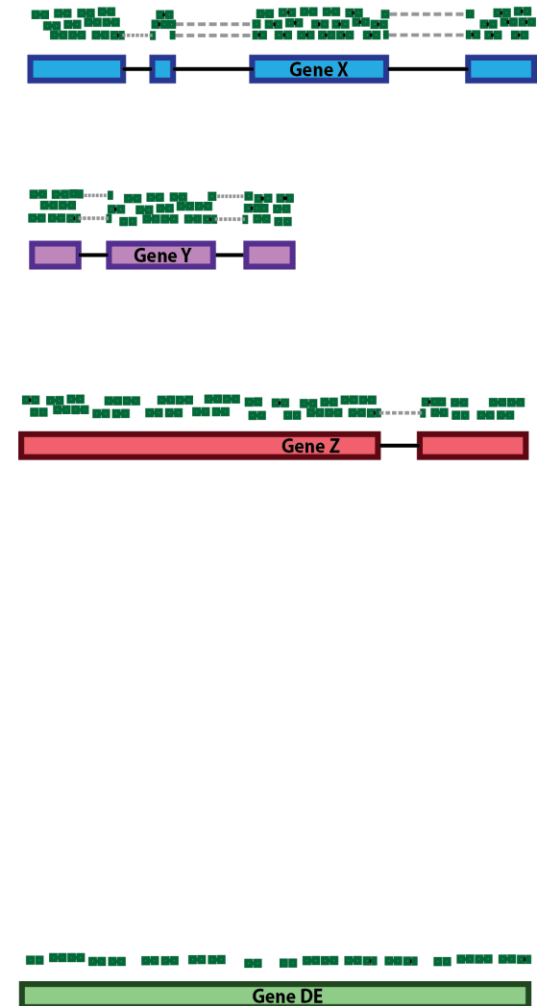
A small set of genes, highly expressed, can bias some normalization methods. It is highly recommended that when comparing two samples with different backgrounds (e.i. different cell types, different genetic backgrounds, etc) the composition of the transcriptome must be taken in account.

In the example, if we were to divide each sample by the total number of counts to normalize, the counts would be greatly skewed by *Gene DE*, which takes up most of the counts for Sample A, but not Sample B. Most other genes for Sample A would be divided by the larger number of total counts and appear to be less expressed than those same genes in Sample B.

Sample A Reads



Sample B Reads



Most popular normalization methods

Counts per million (CPM)

Fragments per kilobase per million reads (FPKM o RPKM)

Transcripts per million reads (TPM)

Trimmed Mean of M-value (TMM - EdgeR)

DESeq's Median of ratios

Normalization methods (I)

Normalization method	Description	Accounted factors	Recommendations for use
CPM (counts per million)	counts scaled by total number of reads	sequencing depth	gene count comparisons between replicates of the same samplegroup; NOT for within sample comparisons or DE analysis
TPM (transcripts per kilobase million)	counts per length of transcript (kb) per million reads mapped	sequencing depth and gene length	gene count comparisons within a sample or between samples of the same sample group; NOT for DE analysis
RPKM/FPKM (reads/fragments per kilobase of exon per million reads/fragments mapped)	similar to TPM	sequencing depth and gene length	gene count comparisons between genes within a sample; NOT for between sample comparisons or DE analysis

Normalization methods (II)

Normalization method	Description	Accounted factors	Recommendations for use
DESeq2's median of ratios [1]	counts divided by sample-specific size factors determined by median ratio of gene counts relative to geometric mean per gene	sequencing depth and RNA composition	gene count comparisons between samples and for DE analysis; NOT for within sample comparisons
EdgeR's trimmed mean of M values (TMM) [2]	uses a weighted trimmed mean of the log expression ratios between samples	sequencing depth, RNA composition, and gene length	gene count comparisons between and within samples and for DE analysis

DESeq2 Normalization: Median of ratios

Since the **differential expression** is tested by comparing counts between sample groups for the same gene, gene length does not need to be accounted for this kind of analysis. However, **sequence coverage and RNA composition need to be considered!**

Step 1: creates a pseudo-reference sample (row-wise geometric mean)

For each gene, a pseudo-reference sample is created that is equal to the geometric mean across all samples.

gene	sampleA	sampleB	pseudo-reference sample
EF2A	1489	906	$\text{sqrt}(1489 * 906) = \mathbf{1161.5}$
ABCD1	22	13	$\text{sqrt}(22 * 13) = \mathbf{17.7}$
...	...	...	...

DESeq Normalization: Median of ratios

Step 2: calculates ratio of each sample to the pseudo-reference

For every gene in a sample, the ratios (sample/ref) are calculated (as shown below). This is performed for each sample in the dataset. Since most genes are not differentially expressed, most genes in each sample should have similar ratios within the sample.

gene	sampleA	sampleB	pseudo-reference sample	ratio of sampleA/ref	ratio of sampleB/ref
EF2A	1489	906	1161.5	$1489/1161.5 = 1.28$	$906/1161.5 = 0.78$
ABCD1	22	13	16.9	$22/16.9 = 1.30$	$13/16.9 = 0.77$
MEFV	793	410	570.2	$793/570.2 = 1.39$	$410/570.2 = 0.72$
BAG1	76	42	56.5	$76/56.5 = 1.35$	$42/56.5 = 0.74$
MOV10	521	1196	883.7	$521/883.7 = 0.590$	$1196/883.7 = 1.35$
...	...	...	...		

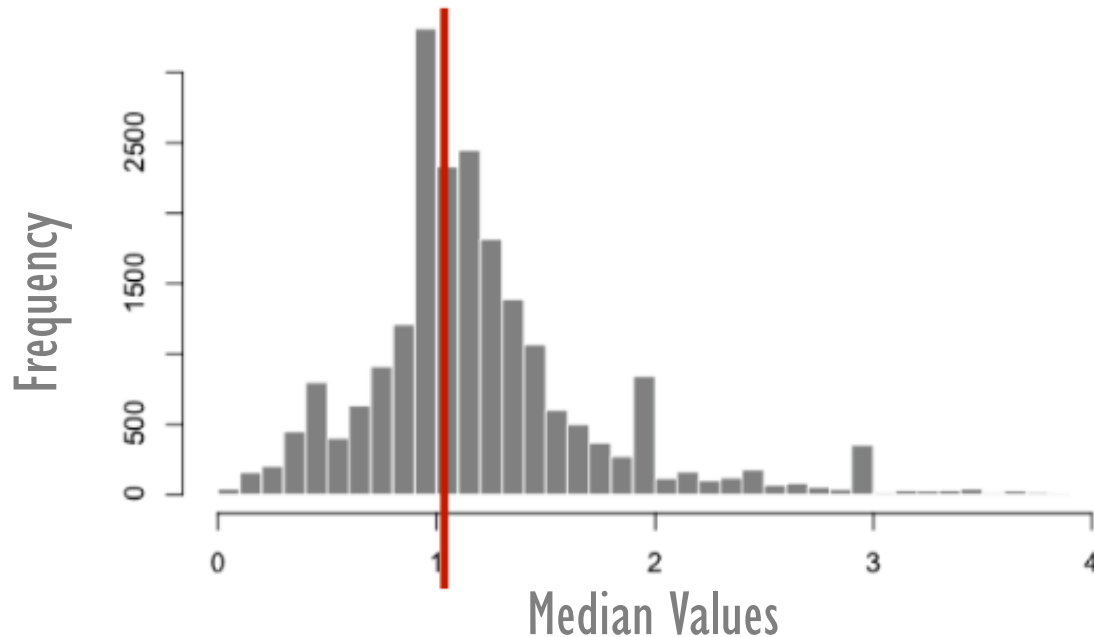
DESeq Normalization: Median of ratios

Step 3: calculate the normalization factor for each sample (size factor)

The median value (column-wise for the step 2 table) of all ratios for a given sample is taken as the normalization factor (size factor) for that sample, as calculated below. Notice that the differentially expressed genes should not affect the median value

```
normalization_factor_sampleA <- median(c(1.28, 1.3, 1.39, 1.35, 0.59))
```

```
normalization_factor_sampleB <- median(c(0.78, 0.77, 0.72, 0.74, 1.35))
```



The histogram shows the frequency and distribution of median values in a sample.

Large outliers (genes differentially expressed) will not be represented by the median values, so, the normalization factors should account for sequencing depth and RNA composition of the sample

DESeq Normalization: Median of ratios

Step 4: calculate the normalized count values using the normalization factor

This is performed by dividing each raw count value in each sample by that sample's normalization factor to generate normalized count values. This is performed for all count values (every gene in every sample).

Raw Counts

gene	sampleA	sampleB
EF2A	1489	906
ABCD1	22	13
...	...	...

Normalized Counts

gene	sampleA	sampleB
EF2A	$1489 / 1.3 = \mathbf{1145.39}$	$906 / 0.77 = \mathbf{1176.62}$
ABCD1	$22 / 1.3 = \mathbf{16.92}$	$13 / 0.77 = \mathbf{16.88}$
...	...	...

Differential Expression Estimation

A right estimation of differentially expressed genes between two conditions is key for understanding phenotypical variations

We should estimate:

The magnitude of differential expression

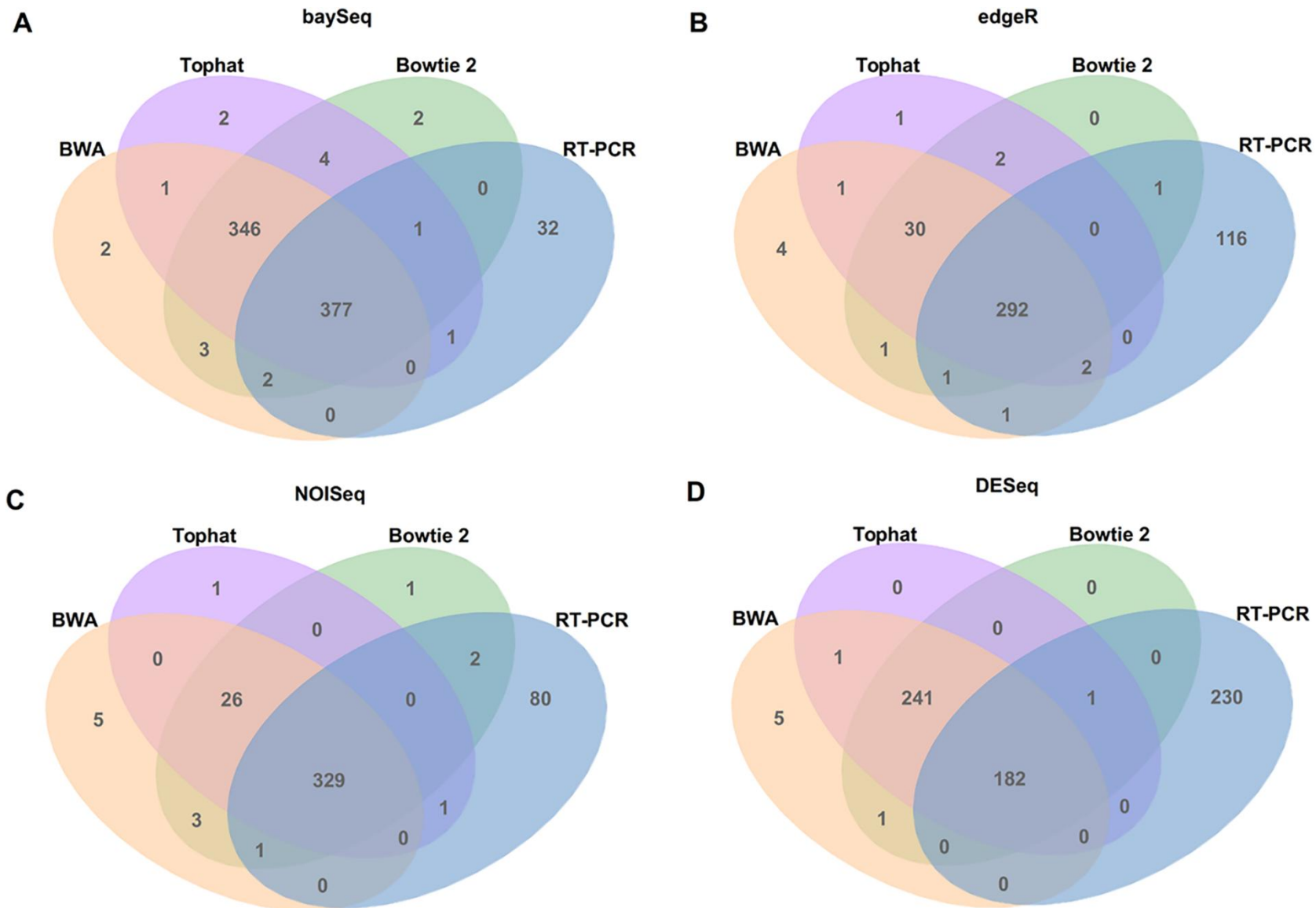
Significance of the differential expression

Methods and software to estimate differential expression:

Name	Version	Normalization	Reference
baySeq	2.4.1	Scaling factors (quantile/ TMM/ total)	[11]
DESeq	1.22.1	DESeq size factors	[12]
EBSeq	1.12.0	DESeq median normalization	[17]
edgeR	3.12.1	TMM/ Upper quartile / RLE / None (all scaling factors are set to be one)	[13]
limma+voom	3.26.9	TMM	[38]
NOIseq	2.14.1	RPKM / TMM / Upper quartile	[15, 39]
SAMseq (samr)	2.0	Based on the read count mean over the null features of data set.	[16]
DESeq2	1.10.1	DESeq size factors	[37]
sleuth	0.29.0	DESeq size factors (with slight modifications)	[35]

<https://doi.org/10.1371/journal.pone.0190152.t002>

Aligner method do not influence DE analysis



Comparison of DE methods with qRT-PCR

Tool	TPR	SPC	PPV	ACC	F_1 measure
edgeR	0.71	0.94	0.90	0.85	0.79
baySeq	0.92	0.40	0.52	0.61	0.66
DESeq	0.44	0.59	0.43	0.53	0.44
NOIseq	0.80	0.95	0.92	0.89	0.86
SAMseq	0.44	0.52	0.39	0.49	0.42
limma+voom	0.81	0.93	0.89	0.88	0.85
EBSeq	0.68	0.55	0.52	0.60	0.59
DESeq2	0.84	0.95	0.92	0.90	0.88
sleuth	0.77	0.54	0.54	0.63	0.64

<https://doi.org/10.1371/journal.pone.0190152.t004>

ACC: Rate of right predictions

NOIseq, Limma+voom and DESeq2 are the programs highly correlating with qRT-PCR results

Comparative table for DE programs

Feature	DESeq2	edgeR	limmaVoom	Cuffdiff
Seq. depth normalization	Sample-wise size factor	Gene-wise trimmed median of means (TMM)	Gene-wise trimmed median of means (TMM)	FPKM-like or DESeq-like
Dispersion estimate	Cox-Reid approximate conditional inference with focus on maximum <i>individual</i> dispersion estimate	Cox-Reid approximate conditional inference moderated towards the <i>mean</i>	squeezes gene-wise residual variances towards the global variance	
Assumed distribution	Neg. binomial	Neg. binomial	<i>log</i> -normal	Neg. binomial
Test for DE	Wald test (2 factors); LRT for multiple factors	exact test for 2 factors; LRT for multiple factors	<i>t</i> -test	<i>t</i> -test
False positives	Low	Low	Low	High
Detection of differential isoforms	No	No	No	Yes
Support for multi-factored experiments	Yes	Yes	Yes	No
Runtime (3-5 replicates)	Seconds to minutes	Seconds to minutes	Seconds to minutes	Hours

LRT: Log-likelihood ratio test

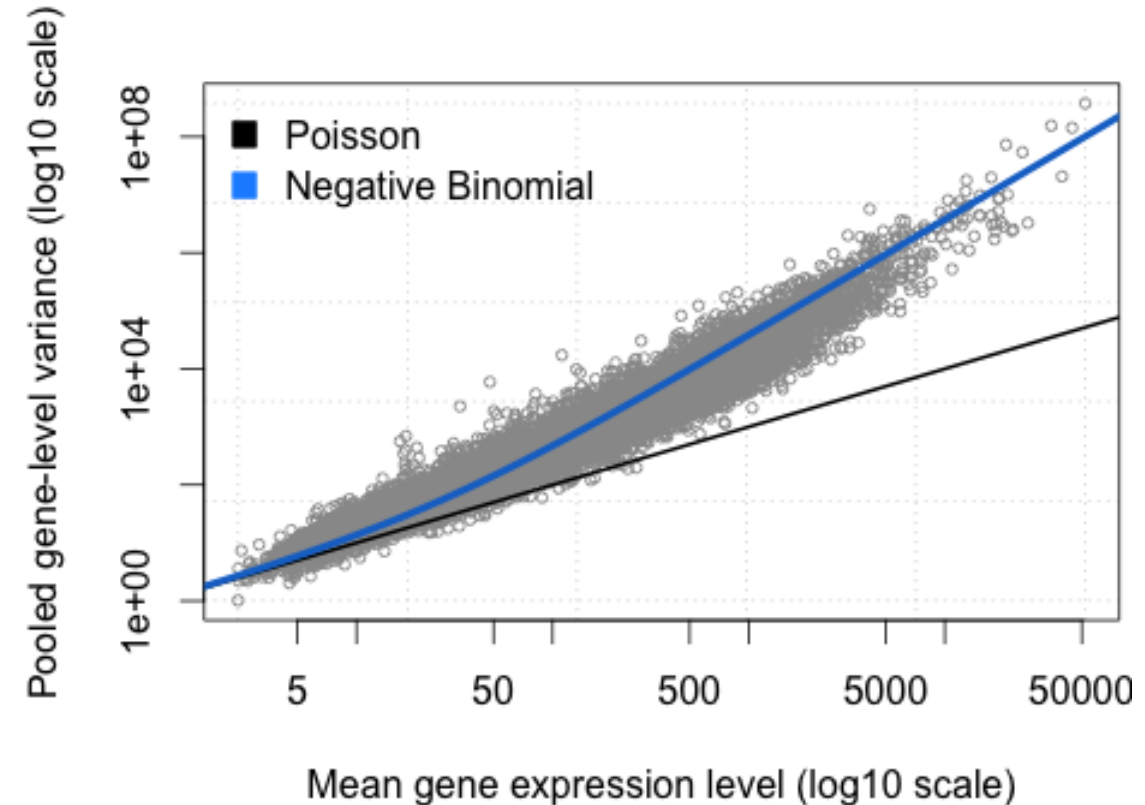
Statistical Models and distribution used by DESeq2

Technical variability of RNA-seq data has been modeled with a Poisson distribution, assuming that variance is equal to the mean of the data

Biological replicates of RNA-seq data can be much more variable than what is expected by Poisson distribution, a phenomenon called “overdispersion”

The statistical model considering overdispersion in RNA-seq data is the **Generalized linear model (GLM)** which uses a negative binomial distribution (allows that the variance exceeds the mean)

DESeq2 and EdgeR combine the estimated dispersion of each gene with an expected dispersion based on all genes. So, for all the reasons explained before we will use DESeq2



I see that...

